1. A student has been saving 10 cent, 20 cent and 50 cent coins in a moneybox. When she opened the box after one month, she found the amount saved is \$15 and the number of 10 cent coins equals the total number of 20 cent and 50 cent coins. She also found that only half as many coins is needed to save the same amount using just 50 cent coins. Find the number of 10 cent, 20 cent and 50 cent coins in the moneybox. [4]

2. Express $4x-3-4x^2$ in the form $-(Ax+B)^2+C$. [2]
Hence solve

$$\frac{(x^2-1)(4x-3-4x^2)}{(x^2-5)} \geq 0,$$

leaving your answer in exact form. [3]

3. A right circular cone with radius 3 cm and height 9 cm is initially full of water. Water is leaking from the circular base of the cone at a rate of 2 $\text{cm}^3\,\text{s}^{-1}$, find the exact rate of change of the depth of water when the depth of water is 6 cm. [5]

4. **(i)** Prove that $\frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{1}{4}\mathrm{e}^{2x}(2x-1)\right] = x\mathrm{e}^{2x}$. [2]

(ii) Hence find $\int x\mathrm{e}^{2x}\ln(2x-1)\,\mathrm{d}x$. [4]

5. The complex number z is given by $z=\cos\theta+\mathrm{i}\sin\theta$, where $-\pi<\theta\leq\pi$. Show that the equation $4+\mathrm{i}\left(4z^3-1\right)=\mathrm{e}^{\mathrm{i}(\pi+3\theta)}$ can be reduced to $z^3=\mathrm{i}$. Hence express each root of $4+\mathrm{i}\left(4z^3-1\right)=\mathrm{e}^{\mathrm{i}(\pi+3\theta)}$ in the form $x+\mathrm{i}y$, where x and y are real numbers. [6]

6. Expand $\left(\dfrac{1-x}{1+x}\right)^n$ in ascending powers of x up to and including the term in x^2. [3]

State the set of values of x for which the series expansion is valid. [1]

Hence find an approximation to the fourth root of $\dfrac{19}{21}$, in the form $\dfrac{p}{q}$, where p and q are positive integers. [3]

7. A hospital patient is receiving a certain drug through a drip at a constant rate of 50 mg per hour. The rate of loss of the drug from the patient's body is proportional to x, where x (in mg) is the amount of drug in the patient at time t (in hour). Form a differential equation connecting x and t and show that

$$x = \frac{1}{k}\left(50 - A\mathrm{e}^{-kt}\right), \qquad [3]$$

where A and k are constants.

Given that $k = \dfrac{1}{20}$.

(i) Find the time needed for the drug in the patient to reach 200 mg, if initially there is no trace of this drug in the patient's body. [2]

(ii) When there is 80 mg of the drug in the patient's body, the drip is disconnected. Assuming that the rate of loss remains the same, find the time taken for the amount of drug in the patient to fall from 80 mg to 20 mg. [3]

8. The functions f and g are defined by

$$\mathrm{f} : x \to -\mathrm{e}^{|3-x|}, \qquad x \in \mathbb{R},$$
$$\mathrm{g} : x \to a-(x-1)^2, \qquad x \in \mathbb{R} \text{ and } a \text{ is a constant.}$$

Sketch the graph of $y = \mathrm{f}(x)$, and show that f does not have an inverse. [2]

(i) The function f has an inverse if its domain is restricted to $x \geq b$. State the smallest possible value of b and define, in similar form, the inverse function f^{-1} corresponding to this domain for f. [4]

(ii) Find the largest value of a such that the composite function $\mathrm{f}^{-1}\mathrm{g}$ exists and state the range of $\mathrm{f}^{-1}\mathrm{g}$ for this value of a. [4]

9. **(a)** A sequence of real numbers $u_1, u_2, u_3, ...$ is defined by $u_1 = 5$ and $u_{n+1} = u_n + 8n + 8$ for all $n \geq 1$. Prove by induction that for every positive integer *n*, $u_n = (2n+1)^2 - 4$. [4]

Show that u_n is a product of two odd numbers. [1]

(b) A sequence of real numbers $u_1, u_2, ...$ is defined by

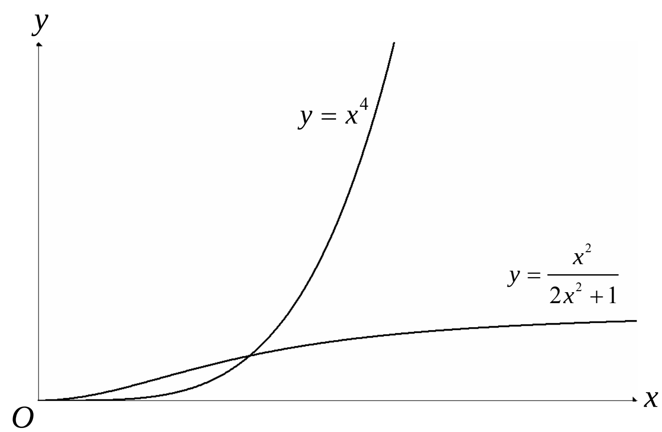
$$u_r = \frac{1}{(2r+1)(2r+3)(2r+5)} \text{ for all } r \geq 1.$$

By taking $\mathrm{f}(r) = \dfrac{1}{(2r+3)(2r+5)}$, express u_r as a difference between $\mathrm{f}(r)$ and $\mathrm{f}(r-1)$. [2]

Evaluate S_n where $S_n = \dfrac{1}{3\cdot5\cdot7} + \dfrac{1}{5\cdot7\cdot9} + ...$ to *n* terms. [3]

Find the limit of S_n as $n \to \infty$. [1]

10. The diagram below shows the graphs of $y = \dfrac{x^2}{2x^2+1}$ and $y = x^4$, for $x \geq 0$. *R* is the region bounded by the *y*-axis, both curves and $y = 1$.



(i) By substituting $y = \frac{1}{2}\sin^2\theta$, find the exact value of $\displaystyle\int_0^{\frac{1}{4}} \sqrt{\frac{y}{1-2y}}\, \mathrm{d}y$. [5]

(ii) Verify that the two curves intersect at the point $\left(\dfrac{1}{\sqrt{2}}, \dfrac{1}{4}\right)$. Hence or otherwise find the exact area of region *R*. [4]

(iii) Find the volume of the solid generated when *R* is rotated through 2π radians about the *y*-axis, giving your answer correct to 3 decimal places. [3]

11. The parametric equations of a curve C are

$$x = -1 - t^2 \quad \text{and} \quad y = \ln(2 - t), \quad t < 2.$$

(i) Sketch the curve C, showing clearly the axial intercepts. [2]

(ii) Find the equations of the normal to the curve at $(-1, \ln 2)$ and the tangent to the curve at $(-5, 2\ln 2)$. [6]

(iii) Find exactly the coordinates of the point of intersection of the tangent and the normal. [2]

(iv) Find, in radians, the acute angle between the tangent and the normal. [2]

12. **(a)** The points A, B, C have position vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ respectively.

(i) Given that non-zero numbers λ, μ are such that $\lambda\mathbf{a} + \mu\mathbf{b} + \mathbf{c} = 0$ and $\lambda + \mu + 1 = 0$. Show that A, B, C are collinear. [2]

(ii) Show that the point P with position vector $\mathbf{p}$ given by $\mathbf{p} = 4\mathbf{a} - 3\mathbf{b}$ lies on BA produced, and find the ratio $PA : PB$. [3]

(b) Referring to the origin O, two planes Π_1 and Π_2 are given by

$$\Pi_1 : \mathbf{r} \bullet \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} = 13 \quad \text{and} \quad \Pi_2 : \mathbf{r} \bullet \begin{pmatrix} 1 \\ 3 \\ 3 \end{pmatrix} = -8.$$

(i) Given that a point A $(1, 7, -10)$ lies on Π_2, show that the perpendicular distance from A to Π_1 is $2\sqrt{21}$. [2]

(ii) Hence or otherwise find $\overrightarrow{OB}$ where B is the image of A when reflected in the plane Π_1. [2]

(iii) Write down the Cartesian equations of both Π_1 and Π_2. [1]

Find a vector equation of the line of intersection of Π_1 and Π_2. [1]

(iv) Find a vector equation of the plane which is the image of Π_2 when Π_2 is reflected in Π_1. [3]